

FELIX BORCK

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RESEARCH SUMMARY

I study how context-sensitive, meaning-like behaviour is causally localized and representationally organized in transformer language models. My work combines philosophy-of-language framing (context dependence, compositionality, vagueness) with intervention-based evaluation and mechanistic interpretability. Two linked projects—*Appearance of Meaning* (behavioural) and *Mechanics of Meaning* (mechanistic)—operationalize the question of whether transformers develop structured semantic competences, and if so, where and in what representational basis those competences reside.

These results provide the basis for testing state-of-the-art mechanistic-interpretability tools, showing that reconstruction fidelity and geometric similarity are useful proxies, but do not certify behavioral preservation under matched intervention. Most recently, I have applied this evidence standard to a safety-relevant control method, testing whether refusal-direction interventions and their sparse representations preserve the behaviour they are claimed to control.

PUBLICATIONS

Borck, F. (2026). *Mechanics of Meaning: Sparse Feature Interventions and the Basis Structure of Contextual Control in Transformers*. Zenodo preprint: zenodo.org/records/18906800

Borck, F. (2026). *The Appearance of Meaning: Context-Dependence and Semantic Competence in Transformer Architectures*. Under review, *Journal of Logic, Language and Information*. Zenodo: zenodo.org/records/18907020

Borck, F. (2026). Geometric Fidelity Does Not Certify Behavioral Preservation. Manuscript / preprint.

Borck, F. (2026). Geometric Fidelity Does Not Certify Causal Preservation in a Refusal-Direction Intervention. Working draft.

Borck, F. (2025). *Context-Dependent Semantics in Large Language Models*. ESSLLI 2025 Student Session. [PhilPapers](https://philpapers.org/).

Borck, F., Löffler, K., Michaels, A. (2018). *Shaping the Future: Green and Sustainable Finance in Germany*. Green and Sustainable Finance Cluster Germany e.V.

Additional manuscripts in preparation on provability theory, three-valued logics, and vagueness in LLMs.

RESEARCH EXPERIENCE

Appearance of Meaning — Behavioural Evaluation & Intervention Pipeline

Independent researcher · 2024–present · [Code](#)

- Designed three governed evaluation suites (DISAMB, CF, COH; 432 controlled JSONL records) for context-dependent semantic competence, with AoM/CPT results reported for GPT-2 and Qwen2.5-{0.5B, 1.5B, 3B}.
- Implemented CPT-style activation patching with context-swap interventions, matched sham baselines, off-target specificity controls, and 1,000-resample bootstrap CIs for intervention-based causal-control analysis.
- Built a unified evaluation/intervention framework covering 7 models across 3 architecture families (GPT-2; Qwen2.5/Qwen3; Llama-3.2), with deterministic runners and run manifests.

Mechanics of Meaning — Mechanistic Interpretability Pipeline

Independent researcher · 2024–present · [Code](#)

- Compared raw-residual vs. SAE-basis interventions at matched patch sites across a 6-layer sweep, with PCA/random baselines and RECON/RESID controls, finding an early-layer (L4) SAE advantage, directional, not yet decisive, consistent with selective projection rather than generic compression.
- Developed top-k feature recovery analysis, showing that 500 features recover about 74.5% of the L4 effect in Gemma 2 2B and probing the concentration structure of context-sensitive representations.
- Built an end-to-end SAE tooling stack for Gemma Scope workflows: loading/calibration, sterility checks, threshold sweeps, feature-space patching, and decomposition metrics, with CPU- and MPS-compatible execution paths.

Geometric Fidelity Does Not Certify Behavioral Preservation - SAE Behavioral-Preservation Tests

Independent researcher · 2024–present · [Code](#)

- Introduced CRR, a matched activation-patching statistic measuring how much of a raw activation-patching effect is preserved by SAE reconstruction patching.
- Found a layer-specific dissociation in Gemma 3 4B: strong reconstruction diagnostics at L4/L8, but CRR 0.621 at L4 versus 0.974 at L8.

- Used PCA, top-k, identity, target-level, and five-layer diagnostics to show that FVU/reconstruction fidelity alone does not certify behavioral preservation.

Refusal-Direction Interventions — Causal Preservation in a Safety-Relevant Control Method

Independent researcher · 2026 · code and manifests available on request

- Replicated the standard single-direction refusal intervention (Arditi et al. pipeline) on Gemma-3-4B-IT under matched instruction-tuned conditions: 0/135 candidate directions passed a pre-registered KL-selectivity gate — a bounded-search null consistent with concurrent independent findings.
- Swept six Gemma Scope SAE layers on Gemma-2-2B-IT (JailbreakBench prompts), measuring how much of the refusal-ablation effect survives SAE write-back: geometric and causal layer rankings dissociate.
- Held-out confirmation on 152 HarmBench prompts: two layers matched in cosine similarity to within 0.002 preserved 0.34 vs. 0.70 of the ablation effect (clipped log-odds; paired bootstrap CI excludes zero).

Philosophical Foundations — Vagueness, Meaning & Formal Semantics

MA research, Goethe University · 2019–2025

- Developed a three-valued semantics (true / false / borderline-unresolved) for vagueness, providing rigorous alternatives to degree-only treatments with applications to evaluation, entity resolution, and policy settings.
- Research on provability operators in modal logic and extensions of Łukasiewicz logic for conditional reasoning; supervised by Prof. André Fuhrmann.

EDUCATION

MA Philosophy · Goethe University Frankfurt · 2018–2025 · Grade: 1.5

Focus: logic, philosophy of language, theories of vagueness. Research with Prof. André Fuhrmann.

BSc Management, Philosophy & Economics · Frankfurt School of Finance & Management · 2014–2018 · Grade: 2.3

Thesis: Climate-related disclosure and the “Tragedy of the Horizon” (grade 1.1). Concentrations in logic, decision theory, and risk management.

TECHNICAL SKILLS

Core: Python, PyTorch, Hugging Face (Transformers + HF Hub), Jupyter

Interpretability: Activation patching / causal tracing, probing, TransformerLens, SAE workflows (loading, calibration, sterility, feature patching, decomposition metrics), bootstrap resampling, matched-control experiments, refusal-direction interventions, jailbreak-benchmark evaluation (JailbreakBench, HarmBench)

Infrastructure: Deterministic runners, manifest-backed provenance, cross-precision validation; CPU / Apple Silicon (MPS) runtimes; AWS (Lambda, DynamoDB, CloudFront, CDK)

ADDITIONAL

Languages: German (native), English (C2), Spanish (B1), French (B1), Arabic (basic)

Industry experience (2018–2024): Frankfurt School UNEP Centre (working student, 2018–2022) and independent consulting (2020–2024) — climate-risk and sustainable-finance work for financial institutions, corporates, and trade associations (transition/physical risk modelling, EU Taxonomy, TCFD, CSRD/ESRS). Self-funded the research program from this work.

DataNXT (co-founder & CEO, 2024–2025): RAG application for regulated financial reporting, with generated statements tagged by provenance and source for review and auditability. Finalist, Goethe University Innovation Prize 2024 and Hessian.AI startup competition; angel investment from Prof. Mark Wahrenburg and the Hessen Ideen Stipendium.

Sapphire Bridge (founder, 2024–present): independent program spanning applied LLM products and mechanistic-interpretability research. Built Voice Post Pilot, a Meta-verified voice-to-content SaaS, solo in ~16 weeks (full-stack, auth, LLM integration, audio/third-party APIs, AWS serverless deployment); runs the interpretability research program above (AoM / MoM / behavioral-preservation).